Exam Date	30/11/2025	Day	Sunday	Time	2:30-3:20
Student Name	[redacted]	Student ID No.	[redacted]	Section	11.30 - 10.30 د.عمر محمد بديل

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** [redacted]

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Multiple Choice	CLO 1,8	20 (67%)			
2	Essay	CLO. 2-7	10 (33%)			
Total			30(100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

D

Question Number	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	D	A	B	A	C	B	C	D	A	A

Question Number	Q11	Q12
Answer	C	D

22

Q1: If an event A satisfies $P(A) = 0.4$ then $P(\bar{A} \mid \Omega)$ equals
A) 0.3 B) 0.7 C) 0.4 D) 0.6 E) None

Q2: For some independent events A, B with $P(A) = P(B) = 0.1$ we have $P(A \cap B)$ equals
A) 0.01 B) 0.1 C) 0.2 D) 0.04 E) None

Q3: Let X be a random variable with zero mean. We have $E(3X^2 + X) =$
A) $2\,Var(X)$ B) $3\,Var(X)$ C) $9\,Var(X)$ D) $4\,Var(X)$ E) None

Q4: For every $x < 0$ we have $u(-x) + 3\delta(x) =$
A) 0 B) 2 C) $undefined$ D) 1 E) None

Q5 : Each ordered pair in $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$ has equal probability. The expected value of XY equals
A) 0.1 B) 0.2 C) 0.25 D) 0.3 E) None

Q6: Let X takes values 0 and 1 with $P(X = 0) = 0.9$ and let Y takes values 1 and 2 with $P(Y = 2) = 0.9$. If X and Y are independent, then $f(1,1) =$
A) 0.1 B) 0.01 C) 0.2 D) 0.02 E) None

Q7: For some continuous function f , if $\int_{-1}^{1} f(x)\,\delta(x)\,dx = 1$, then $f(0) =$
A) 0 B) -1 C) 1 D) -2 E) None

Q8: For the joint PDF $f(x,y) = x$ when $0 < x < \sqrt{2}$, $0 < y < 1$ and zero otherwise. We have $f_X(x) =$
A) 1 B) y C) $\sqrt{2}y$ D) x E) None

Q9: For a continuous random variable X with $P(1 < X < 2) = 0.3$, Then $P(0 < X < 2) - P(0 < X < 1) =$
A) 0.3 B) 0.4 C) 0.2 D) 0.1 E) None

Q10 : For two random variables X, Y with $Cov(X,Y) = 1$, $Cov(\pi - eX, e + \pi Y) =$
A) $-e\pi$ B) e C) $e\pi$ D) π E) none

Q11: Let X, Y be two independent random variables with $Var(X) = Var(Y)$ Then $Corr(X, Y) =$

A) 1 B) 1 C) 0 D) 3 E) None

Q12: A discrete random variable X with CDF $F(x) = a\,u(x+2) + a\,u(x-1)$

Then the constant a equals

A) 0 B) 0.25 C) 0.2 D) 0.5 E) None

X, y independed

$\forall$ $Cov(X,y) = EXY - EXEy$

$= EXEy - EXEy$

$= 0$

$\int_0^x a\,du\,x+2 + \int_0^x a\,(u+1)$

$\int_0^x 2a = 1$

$(x-0)$

$2a = 1$

النموذج (1)

Q1:- Find $P(\bar{A}|\Omega)$

$P(A) = 0.4$

$P(\bar{A}) = 1 - 0.4 = 0.6$

$P(\Omega) = 1$

$$P(\bar{A}|\Omega) = \frac{P(\bar{A}) * P(\Omega)}{P(\Omega)} = 0.6 \longrightarrow D$$

Q2:-

A, B Independent

$P(A) = P(B) = 0.1$

$P(A \cap B) = ??$

$P(A \cap B) = P(A) * P(B)$ (Independen)

$= 0.1 * 0.1 = 0.01 \longrightarrow A$

Q3:-

Zero mean

$\mu \rightarrow EX = 0$

Find

$E(3X^2 + X)$

Sol:-

$E(3X^2 + X) = 3EX^2 + EX = 3EX^2 = 3 \text{var} X \longrightarrow B$

$\text{var} X = EX^2 + (EX)^2$

$\text{var} X = EX^2 + (0)^2$

$\text{var} X = EX^2$

Q4:

$x \geq 0$, Find

$u(-x) + 3\delta(x) \longrightarrow$

Sol:-

$u(-x)$, $x \geq 0$

$u(-x) = 1$

$3\delta(x) = 0$

$u(-x) + 3\delta(x) = 1 + 3(0) = 1 \longrightarrow D$

Q5:- $R_{xy} = \{(0,0), (0,1), (1,0), (1,1)\}$ has equal probability

$$E_{xy} = \sum_{R_x} \sum_{R_y} xy\, f(x,y) = (0)(0) f(0,0) + (0)(1) f(0,1) + (1)(0) f(1,0) + (1)(1) f(1,1)$$

$= 0 + 0 + 0 + (1)(1)(0.25)$

$= 0.25 \longrightarrow C$

$$\sum_{R_x} \sum_{R_y} f(x,y) = 1$$

same

$f(0,0) + f(0,1) + f(1,0) + f(1,1) = 1$

$4f(1,1) = 1$

$f(1,1) = 1/4 = 0.25 \Rightarrow f(0,0) = f(0,1) = f(1,0)$

Q6:-

X, Y Independent

$X = \{0, 1\}$
$Y = \{1, 2\}$
$P(X=0) = 0.9$
$P(Y=2) = 0.9$

Find $f(1,1) \longrightarrow$ sol $F(1,1) = f_x(1) * f_y(1) \rightsquigarrow$ Independent

$P(X=1) = 1 - P(X=0) = 1 - 0.9 = 0.1$

$P(y=1) = 1 - P(y=2) = 1 - 0.9 = 0.1$

$f(1,1) = f_x(1) * f_y(1)$

$= 0.1 * 0.1 = 0.01 \longrightarrow B$

Q7:-

f continuous function

$\int_{-1}^{1} f(x)\,\delta(x) = 1$, Find $f(0)$

$\int_{-1}^{-1} f(x)\,\delta(x) = f(0)$

$0 \in [-1, 1]$

$f(0) = 1 \longrightarrow C$

Q8:- Joint pdf

$f(x,y) = x$, $0 < x < \sqrt{2}$, $0 < y < 1$

Find $f_x(x) = ??$

Sol:-

$f_x(x) = \int_{R_y} f(x,y)\,dy$

$= \int_0^1 x\,dy = x \times [y]_0^1 = x(1-0) = \underline{x} \longrightarrow D$

or

$= x(1-0) = x$

Q9:- X continuous r.v

$P(1 < X < 2) = 0.3$

Find $= F(2) - F(1) = 0.3$

$P(0 < X < 2) - P(0 < x < 1)$

Sol:-

$P(a < X < b) = F(b) - F(a) \rightsquigarrow$ continuous

$P(0 < X < 2) - P(0 < X < 1)$

$= F(2) - F(0) - [F(1) - F(0)]$

$= F(2) - F(0) - F(1) + F(0)$

$= F(2) - F(1) = 0.3 \longrightarrow A$

Q10:-

X, Y two r.v

Cov(X,Y) = 1

Find

Cov(π - eX, e + πY)

Cov(π - eX, e + πY)

= (a)(c) Cov(X,Y)

= (-e)(π)(1) = -eπ ---> A

Q11:-

X, Y two Independent r.v

Var(X) = Var(Y)

Find

Corr(X,Y) = ??

Sol:-

X, Y ---> Independent

Cov(X,Y) = 0 , Corr(X,Y) = 0 ---> C

↳ $= \frac{Cov(X,Y)}{\sigma_x \sigma_y} = \frac{0}{\sigma_x \sigma_y} = 0$

Q12:-

X is a discrete r.v with CDF

F(X) = a(x+2) + a(x-1)

Find a ??

Sol:-

F(∞) = 1

a(x+2) + a(x-1) = 1

2a = 1

a = 1/2 = 0.5 ---> D

Q13:- Poisson distribution

Sol:-

→ $X \sim P(m)$, m = 3

$X \sim P(3)$

Find P(X=4)

$P(X=x) = \frac{e^{-m} m^x}{x!}$

$P(X=4) = \frac{e^{-4}(3)^4}{4!} = \frac{81e^{-4}}{24}$ ✓

Q14:- Binomial distribution

$X \sim B(n,p)$, $X \sim B(10, 0.8)$

n = 10

p = 80% = 0.8 , $q_{fail} = 1 - 0.8 = 0.2$

succes

Find P(X ≥ 1) = ??

P(X ≥ 1) = 1 - P(X < 1)

= 1 - P(X=0)

$= 1 - \binom{10}{0}(0.8)^0(0.2)^{10}$ ✓

$P(X=x) = \binom{n}{x} p^x q^{n-x}$

Exam Date	30/11/2025	Day	Sunday	Academic Year	2025/2026
Student Name	[redacted]	Student ID No.		Time	2:30-3:20
				Section	د. عمر بدير 10:30/11:30 خ ش ز

GENERAL INSTRUCTIONS

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
1	Multiple Choice	CLIO.1,8	20 (67%)			
2	Essay	CLIO, 2-7	10 (33%)			
Total			30(100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

Question Number	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Answer	B	D	A	B	C	D	A	B	D	C

Question Number	Q11	Q12
Answer	A	B

24

Q1: If an event A satisfies $P(A) = 0.3$ then $P(\overline{A} \mid \Omega)$ equals
A) 0.3 B) 0.7 C) 0.4 D) 0.6 E) None

Q2: For some independent events A, B with $P(A) = P(B) = 0.2$ we have $P(A \cap B)$ equals
A) 0.01 B) 0.1 C) 0.2 D) 0.04 E) None

Q3: Let X be a random variable with zero mean. We have $E(2X^2 + X) =$
A) $2\,Var(X)$ B) $3\,Var(X)$ C) $9\,Var(X)$ D) $4\,Var(X)$ E) None

Q4: For every $x < 0$ we have $2u(-x) + 3\delta(x) =$
A) 0 B) 2 C) $undefined$ D) 1 E) None

Q5 : Each ordered pair in $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$ has equal probability. The expected value of XY equals
A) 0.1 B) 0.2 C) 0.25 D) 0.3 E) None

Q6: Let X takes values 0 and 1 with $P(X = 0) = 0.9$ and let Y takes values 1 and 2 with $P(Y = 2) = 0.8$. If X and Y are independent, then $f(1,1) =$
A) 0.1 B) 0.01 C) 0.2 D) 0.02 E) None

Q7: For some continuous function f , if $\int_{-1}^{1} f(x)\,\delta(x)dx = 0$, then $f(0) =$
A) 0 B) 1 C) -1 D) -2 E) None

Q8: For the joint PDF $f(x,y) = y$ when $0 < x < 1$, $0 < y < \sqrt{2}$ and zero otherwise. We have $f_Y(y) =$
A) 1 B) y C) $\sqrt{2}y$ D) x E) None

Q9: For continuous random variable X with $P(1 < X < 2) = 0.1$, Then $P(0 < X < 2) - P(0 < X < 1)$ equals
A) 0.3 B) 0.4 C) 0.2 D) 0.1 E) None

Q10 : For two random variables X, Y with $Cov(X,Y) = 1$, $Cov(\pi - eX, e - \pi Y) =$
A) $-e\pi$ B) e C) $e\pi$ D) π E) none

Q11: Let X, Y be two independent random variables with $Var(X) = Var(Y)$ Then $Corr(X, Y) =$

A) 0 B) 1 C) 2 D) 3 E) None

Q12: A discrete random variable X with CDF

$$F(x) = a\,u(x+2) + 3a\,u(x-1)$$

Then the constant a equals

A) 0 B) 0.25 C) 0.2 D) 0.5 E) None



Q13: A call center receives an average of 3 calls per hour. Calculate the probability that exactly four calls arrive in a given hour!

$x=4$ $m=3$ ← Poisson

$x \sim P(m)$

$x \sim P(3)$

So we are $\frac{e^{-m} \cdot m^x}{x!}$

$P(x) = \frac{e^{-3} \cdot (3)^4}{4!} = 0.168$

Q14: A machine makes items, and each item has an 80% chance of being good. If we check 10 (n) items, what is the probability that at least one item is good?

$x \sim B(n \cdot P)$ $x \sim B(10, \frac{4}{5})$ $P = 80\% = \frac{4}{5}$

$q = 1 - P = \frac{1}{5}$

$P(x) = \binom{n}{x} P^x q^{n-x}$

$= P(x \geq 1)$

$1 - P(x \leq 1)$

$1 - P(x=0)$

$1 - \binom{n}{x} P^x q^{n-x}$

$1 - \left(\binom{10}{0} P^0 q^{10-0}\right)$

$1 - \left(1 \left(\frac{4}{5}\right)^0 \left(\frac{1}{5}\right)^{10}\right) = 0.999 = 1 - 0.2^{10}$

$P(x \geq 1) = 0.999$

النموذج (2) :-

Q1:-

Find $P(\bar{A}|\Omega)$

$P(A) = 0.3$

$P(\bar{A}) = 1 - 0.3 = 0.7$ (P(A))

$P(\Omega) = 1$

Soli-

$$P(\bar{A}|\Omega) = \frac{P(\bar{A}) * P(\Omega)}{P(\Omega)} = \underline{0.7} \longrightarrow B$$

Q2:-

A, B Independent

$P(A) = P(B) = 0.2$

$P(A \cap B) = ??$

Soli-

$P(A \cap B) = P(A) * P(B)$

$= 0.2 * 0.2$

$= 0.04 \longrightarrow D$

Q3:-

mean

$E[X] = \mu = 0$

Find

$E(2x^2 + x)$

Soli-

$Ex = 0$, $E(2x^2 + x) = 2Ex^2 + \cancel{Ex}^{\to 0}$

$Var X = Ex^2 + \cancel{(Ex)^2}^{\to 0}$

$\therefore 2 Var X \longrightarrow A$

Q4:-

$x < 0$ Find $2u(-x) + 3\delta(x)$

Soli-

$2u(-x) + 3\delta(x) = 2(1) + (0)(3) = 2 \longrightarrow B$

Q5:- [redacted]

$\sum\sum f(x,y) = 1$

نفس حل النموذج السابق

same Prob.

$f(0,0) + f(1,0) + f(0,1) + f(1,1) = 1$

$4\, f(1,1) = 1/4 = f(0,1) = f(1,0) = f(0,0)$

Ex. $\sum\sum xy\, f(x,y) = 0.25 \longrightarrow C$

Q6:-

$X = \{0, 1\}$

$Y = \{1, 2\}$

$P(X=0) = 0.9$

$P(X=2) = 0.8$

X, Y independent

$f(1,1) = ??$

$f_{(1,1)} = f_x(1) * f_y(1)$ ⇝ independent

$P(X=1) = 1 - P(X=0) = 0.1$

$P(X=1) = 1 - P(X=2) = 0.2$

$= 0.1 * 0.2 = 0.02 \longrightarrow D$

Q7: f is continuous function

$$\int_{-1}^{1} f(x)\,\delta(x) = 0$$ Find $f(0)$

Solv

$x_0 = 0 \in [-1, 1]$ ✓

$$f(0) = \int_{-1}^{1} f(x)\,\delta(x) = 0 \longrightarrow A$$

Q8 Pdf Joint $f(x,y) = y$, $0 < x < 1$, $0 < y < \sqrt{2}$

Find $f_y(y)$

Solv

$$f_y(y) = \int_{Rx} f(x,y)\,dx = \int_0^1 y\,dx = y(1-0) = y \longrightarrow B$$

Q9:-

X continuous r.v $P(1 < X < 2) = 0.1$

Find $P(0 < X < 2) - P(0 < X < 1) = ?$

Solv

$P(a < x < b) = F(b) - F(a)$

$P(0 < X < 2) - P(0 < X < 1) = F(2) - F(0) - [F(1) - F(0)]$

$= F(2) - F(0) - F(1) + F(0) = F(2) - F(1) = 0.1 \longrightarrow D$

Q10:-

$Cov(x,y)=1$

Find $Cov(\pi - ex, e - \pi y)$

Sol:-

$Cov(\pi - ex, e - \pi y) = (-e)(-\pi)\,Cov(x,y) = e\pi \longrightarrow C$

Q11:- X, Y Independent r.v , Find Corr(x,y)

$Corr(x,y) = 0 \longrightarrow A$

Q12:-

$F(x) = a\,u(x+2) + 3a(x-1)$, Find a

Sol:-

$F(\infty) = 1$,

$a + 3a = 1$

$4a = 1$

$a = 1/4 = 0.25 \longrightarrow B$

Q13:- Poisson distribution

$X \sim P(M)$, $M = 3$

$X \sim P(3)$

$P(X=x) = \dfrac{e^{-m} * m^{x}}{x!}$

$P(X=4) = \dfrac{e^{-3} * 4^{3}}{4!}$

$= \dfrac{81\,e^{-4}}{24}$

Q14:- Binomial distribution

$X \sim B(n,p)$, $n = 10$

$X \sim B(10, 0.8)$, $p = 80\% = 0.8$

$P(x \geq 1) = 1 - P(X=0)$, $q = 1 - 0.8 = 0.2$

$= 1 - \binom{10}{0}(0.8)^{0}(0.2)^{10}$